

BINF 5506 Bioinformatics and Cloud Computing for Advanced Data Integration and Management Critical Path

Week#	Class Type	Topics, Readings, and Activities
Week 1	LESSON	Course Administration Learning Outcomes Critical Path Deployment Models Cloud Computing in Bioinformatics Amazon Web Services Google Cloud Microsoft Azure Amazon Web Services Learner Lab Setup
	LAB	
Week 2	LESSON	Accessing AWS Services AWS Learner Lab Data Centers Regions Availability Zones AWS Virtual Private Cloud (VPC) AWS Key Pairs AWS Elastic Compute Cloud (EC2) Setup
	LAB	Assignment 1 Release
Week 3	LESSON	Elastic Compute Cloud (EC2) EC2 Instance Connect (Web CLI) Root User (Super User) Linux Package Management (apt) Anaconda Installation EC2 Lifecycle Amazon Machine Images (AMI) Elastic IP Addresses
	LAB	
Week 4	LESSON	Elastic Block Storage OpenSSH VSCode Remote SSH Web Servers Apache
	LAB	Assignment 1 Due
Week 5	LESSON	AWS Simple Storage Service (S3) S3 Use Cases S3 Versioning S3 Storage Classes S3 Lifecycle Choosing an AWS Region

BINF 5506 Bioinformatics and Cloud Computing for Advanced Data Integration and Management Critical Path

		AWS SDK for Python (boto3) DrawIO
	LAB	Test 1 Assignment 2 Release
Week 6	LESSON	Git Basics GitHub Snakemake Introduction Rule Syntax Input/Output Management Dependency Resolution Wildcard Usage Parallel Execution IAM
	LAB	
Week 7	LESSON	Docker Introduction Containers Docker Files Docker Compose Docker Image Tagging and Versioning AWS Elastic Container Registry
	LAB	Assignment 2 Due / Assignment 3 Release
Week 8	LESSON	AWS Elastic Container Service (ECS) AWS ECS Task Definitions Fargate vs EC2 Launch Types AWS Batch AWS Batch Job Definitions AWS Batch Queues AWS Monitoring with CloudWatch
	LAB	
Week 9	LESSON	Serverless Computing Fundamentals AWS Managed vs. Unmanaged Services AWS Lambda AWS Lambda Function Deployment Event-Driven Architectures
	LAB	Assignment 3 Due / Assignment 4 Release
Week 10	LESSON	AWS SageMaker Notebook Instances Training Jobs & Tuning Model Deployment & Endpoints Pipelines & Automation Debugging & Monitoring
	LAB	

**BINF 5506 Bioinformatics and Cloud Computing for Advanced Data Integration
and Management Critical Path**

Week 11	LESSON	AWS Step Functions State Machine Design Task Orchestration Error Handling Strategies Conditional Branching
	LAB	
Week 12	TEST	Test 2
	LAB	Assignment 4 Due

SUMMARY	WEIGHTAGE
Assignments (4)	70
Tests (2)	30
TOTAL	100

Note: This critical path may change as resources and circumstances require.